

Sampling Design Governs the Aggregate Sweep Ratio: Exhaustive Per-Node Sweeps on a Reactive Graph Converge to $2\times$, Universally

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Abstract

A companion paper proves that on a reactive computational graph – one whose invalidation provably touches exactly the downstream cone of a mutated node – the aggregate speedup of an exhaustive sweep over independent full recomputations is not a universal constant: under a depth-stratified sampling protocol (a fixed number of candidate sites per layer, independent of the layer’s computational weight), the ratio converges to $(q+2)/(q+1)$ or $q+2$ in the Karamata index q of the network’s cost-by-depth profile, recovering 2 only in the depth-uniform case. This note proves that this q -dependence is entirely an artifact of that sampling protocol, not a structural property of the graph. Under the sampling protocol an exhaustive sweep actually uses in practice – every computable node is a candidate site, so a heavier layer contributes proportionally more sites – the aggregate ratio converges to exactly 2 for *every* cost profile and either orientation, at an explicit rate (governed by depth alone, $\Theta(1/L)$, when layers have genuine internal width; faster, $\Theta(1/N)$, on a fully sequential chain), via an exact pair-counting identity that requires no regular-variation hypothesis at all: for layerwise-antichain graphs, $2D(L) = N(L)^2 - \|w\|_2^2$ for downstream-cone total $D(L)$ and per-layer weights w . On a fully sequential chain the identity degenerates further to $D(L) = \binom{N(L)}{2}$ exactly, at every finite depth, independent of the cost profile entirely. We verify both identities at zero numerical tolerance on real reactive graphs, under the same four Karamata profiles the companion paper’s aggregate-ratio theorem was tested against, and show that a single macroscopic layer (violating the theorem’s only hypothesis) restores a different, also-computable limit of $8/3$.

1 Introduction

A companion paper, *Cost Accounting for Reactive Computational Graphs* [3], proves that on a reactive graph engine – a persistent computational graph whose mutation invalidation is provably confined to the exact downstream cone of the mutated node, a guarantee established in a second companion paper, *Exact Network Surgery* [2] – the cost of an exhaustive site-by-site sweep (patch every candidate site, restore, move to the next) is governed by a single number: the Karamata index q of how computational weight is distributed across depth. The aggregate speedup over independent full recomputations converges to $(q+2)/(q+1)$ when weight concentrates near the output, to $q+2$ when it concentrates near the input, and to a universal 2 only in the special case $q=0$ (uniform weight per layer).

That theorem samples a *fixed number of sites per layer*, independent of the layer’s own weight – the natural model for a sweep that probes one representative site per depth (one residual-stream position, say). A different, equally real sweep protocol – probing every individual computable unit as its own candidate site, the natural model for a neuron- or edge-level exhaustive ablation – samples sites *in proportion to* each layer’s weight: a layer twice as heavy hosts twice as many sites. This note shows that under this second protocol, the q -dependence vanishes entirely: the

aggregate ratio is exactly 2 in the limit, for every profile, regardless of orientation. The two theorems are not in tension; they are a matched pair that isolates exactly what the aggregate ratio actually measures – a readout of the *site-sampling design*, crossed with the cost profile, not a structural invariant of the graph by itself.

Contributions.

1. A definition of the *exhaustive per-node sweep* sampling regime, contrasted explicitly with the depth-stratified regime of the companion aggregate-ratio theorem (Section 3).
2. A theorem giving, for this regime: an exact sandwich on the aggregate ratio at every finite depth (no asymptotic hypothesis at all), a universal limit of 2 under a hypothesis strictly weaker than – and implied by – the companion theorem’s Karamata hypothesis, the exact convergence rate in both the layerwise-antichain case ($\Theta(1/L)$) and the totally ordered case ($\Theta(1/N)$, faster whenever $q > 0$), and an orientation-invariance property with no analogue in the depth-stratified regime (Theorem 2).
3. A necessity result: a single asymptotically macroscopic layer breaks the universal limit, converging instead to a different, exactly computable constant (Remark 2).
4. Zero-tolerance numerical verification on real reactive graphs in the reference implementation, *NeuroDSL* [1], under the same four cost profiles the companion paper’s aggregate-ratio theorem was validated against, plus a fully sequential-chain confirmation and a deliberate hypothesis-violating construction (Section 4).

2 Preliminaries

We reuse, without re-proving, two results from the companion papers. First (*Exact Network Surgery* [2]), a computational graph is a finite DAG $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \text{op}, \theta)$; a reactive invalidation algorithm, seeded at a mutated node s , marks invalid exactly the *downstream cone* $\mathcal{V}_s^+ = \{v \in \mathcal{V} \mid \text{a path of length } \geq 1 \text{ from } s \text{ to } v\}$, at cost $O(|\mathcal{V}_s^+| + |\mathcal{E}_s^+|)$ – the *structural locality theorem*. Second (*Cost Accounting* [3]), a *layered cost model* of depth L assigns to layer $j \in \{1, \dots, L\}$ a computational weight $w_j > 0$ (its node count) with $N(L) = \sum_j w_j$, and stipulates that a site’s downstream cone coincides, up to an $O(1)$ per-site remainder that vanishes in every ratio below, with the suffix $\sum_{j>i} w_j$ of the layer i it belongs to.

Definition 1 (Depth-stratified sweep, recalled). *The depth-stratified sweep of [3] places S candidate sites per layer, independent of w_j , and its aggregate ratio is $\rho(L) = SL \cdot N(L) / S \sum_{i=1}^L \sum_{j>i} w_j$.*

Theorem 1 (Aggregate sweep ratio, recalled [3]). *If $w_j \sim \ell(j)j^q$ for ℓ slowly varying and $q \geq 0$ (Karamata), then $\rho(L) \rightarrow (q+2)/(q+1)$ for this output-heavy profile, and $\rightarrow q+2$ for the reversed (input-heavy) profile $w_j \sim \ell(L+1-j)(L+1-j)^q$; both recover 2 at $q = 0$.*

3 The Exhaustive Per-Node Sweep

Definition 2 (Exhaustive per-node sweep). *In the layered cost model, assume the weights w_j are integers and layer j consists of w_j unit-cost nodes. The exhaustive per-node sweep takes every node as a candidate site, so layer j hosts w_j sites – in contrast with the depth-uniform placement of Definition 1. With $\mathcal{V}_v^+$ the downstream cone of node v (every node of every deeper layer, together with v ’s descendants inside its own layer, up to the same vanishing per-site remainder as above), write*

$$D(L) = \sum_v |\mathcal{V}_v^+|, \quad \rho_{\text{node}}(L) = \frac{N(L)^2}{D(L)}$$

for the ratio of $N(L)$ independent full recomputations to the cost of exhaustively patching every node, each restored via the structural locality theorem's exact cone.

Theorem 2 (Universality of the exhaustive per-node ratio). *Let $\|w\|_2^2 = \sum_{j=1}^L w_j^2$ and $N = N(L)$. For every weight profile (w_j) :*

(i) (Exact sandwich.) $D(L)$ counts exactly the ordered comparable node pairs, and

$$\frac{N^2 - \|w\|_2^2}{2} \leq D(L) \leq \binom{N}{2}, \quad \text{hence} \quad \frac{2N}{N-1} \leq \rho_{\text{node}}(L) \leq \frac{2}{1 - \|w\|_2^2/N^2},$$

the left bound on D (right bound on ρ_{node}) attained iff every layer is an antichain (no two nodes of the same layer comparable), and the right bound on D iff every layer is internally totally ordered – in which case the comparability relation among nodes is a single total order (as realized, e.g., by a fully sequential chain) and $\rho_{\text{node}}(L) = 2N/(N-1)$ exactly, at every finite L , for every profile, with no asymptotic hypothesis whatsoever.

(ii) (Universal limit.) If $\max_j w_j/N(L) \rightarrow 0$ – no single layer carries an asymptotically positive fraction of the total weight – then $\rho_{\text{node}}(L) \rightarrow 2$ for every profile and either orientation. This hypothesis is strictly weaker than, and implied by, Theorem 1's Karamata hypothesis: under either profile there, $\|w\|_2^2/N^2 \sim (q+1)^2/((2q+1)L) \rightarrow 0$ for every $q \geq 0$.

(iii) (Rate.) In the totally ordered case, $\rho_{\text{node}} = 2 + 2/(N-1)$ exactly, i.e. $\Theta(1/N)$ – which, under a Karamata profile, is $\Theta(1/L^{q+1})$ since $N \sim Lw_L/(q+1)$, strictly faster than $1/L$ whenever $q > 0$. In the layerwise-antichain case, $\rho_{\text{node}} = 2 + \frac{2(q+1)^2}{(2q+1)L} (1 + o(1))$ under either Karamata profile of Theorem 1: here the rate is genuinely $\Theta(1/L)$, governed by depth alone, since intra-layer width does not shrink the cone in this construction. The two cases coincide only at $q = 0$, where $N \sim L$.

(iv) (Orientation invariance.) At every finite L , both bounds in (i) – and ρ_{node} itself in the antichain and totally ordered cases – are invariant under profile reversal $w_j \mapsto w_{L+1-j}$, unlike the asymmetric pair $(q+2)/(q+1)$ vs. $q+2$ of Theorem 1.

Proof. $D(L)$ counts ordered pairs (v, u) with $u \in \mathcal{V}_v^+$, i.e. comparable pairs. Decompose by layers. *Inter-layer:* every node of layer i has every node of every layer $j > i$ in its cone, contributing $\sum_{i < j} w_i w_j = \frac{1}{2}(N^2 - \|w\|_2^2)$, by expanding $N^2 = (\sum_j w_j)^2 = \|w\|_2^2 + 2 \sum_{i < j} w_i w_j$. *Intra-layer:* layer i contributes its own count D_i of comparable ordered pairs, $0 \leq D_i \leq \binom{w_i}{2}$, extremes attained by an antichain and a total order respectively. Summing,

$$\frac{N^2 - \|w\|_2^2}{2} \leq D(L) \leq \frac{N^2 - \|w\|_2^2}{2} + \sum_i \frac{w_i(w_i - 1)}{2} = \frac{N^2 - N}{2} = \binom{N}{2},$$

proving (i); the per-site remainders of the layered cost model total $o(N^2)$ and vanish in the ratio. For (ii), $\|w\|_2^2/N^2 \leq (\max_j w_j/N) \cdot (\sum_j w_j/N) = \max_j w_j/N \rightarrow 0$ and $N \geq L \rightarrow \infty$; both bounds of (i) converge to 2, so $\rho_{\text{node}} \rightarrow 2$ by squeezing. Under a Karamata profile, w_j^2 is regularly varying of index $2q$, so Karamata's theorem gives $\|w\|_2^2 \sim Lw_L^2/(2q+1)$, and with $N \sim Lw_L/(q+1)$ (as in Theorem 1's proof) the ratio is $\sim (q+1)^2/((2q+1)L)$; the reversed profile leaves N and $\|w\|_2^2$ unchanged, both being symmetric functions of the multiset $\{w_j\}$. (iii) follows from $2/(1-x) = 2 + 2x + O(x^2)$ at $x = \|w\|_2^2/N^2$, and from (i) in the totally ordered case. (iv) is immediate: N and $\|w\|_2^2$ are invariant under $j \mapsto L+1-j$, and in the two extreme cases ρ_{node} is a function of these two quantities alone. $\square$

Remark 1 (Sampling design, not graph structure). *Theorem 1 and Theorem 2 are a matched pair: any aggregate ratio of this kind equals $1/\mathbb{E}[U]$ for U the downstream-weight fraction of a*

site drawn uniformly among the sites, and the two theorems differ only in the law this induces on depth. Depth-uniform placement draws depth uniformly, and $\mathbb{E}[U]$ reads the cost profile directly – the q -dependent limits of Theorem 1. Per-node placement draws depth size-biased by weight, and then, writing p for the (continuum limit of the) normalized weight density and $\bar{P}(t) = \int_t^1 p$ its survival function, $\mathbb{E}[U] = \int_0^1 p(t)\bar{P}(t) dt = \frac{1}{2}[\bar{P}(0)^2 - \bar{P}(1)^2] = \frac{1}{2}$ identically, for every density p ; discretely, this continuum identity is nothing but the pair-counting identity $\sum_{i < j} w_i w_j = (N^2 - \|w\|_2^2)/2$ used in the proof above, an algebraic fact that requires no regular variation and holds at every finite L . Neither regime is an artifact of the other: standard activation-patching protocols place a fixed set of sites per layer (one per residual-stream position, say) – the depth-stratified regime of Theorem 1 – while neuron- or edge-level exhaustive ablations are the per-node regime of Theorem 2. The universal constant 2 is the signature of size-biased (genuinely exhaustive) placement, approached from above at a rate governed by depth ($\Theta(1/L)$) when layers have genuine width, or faster ($\Theta(1/N)$) on a fully sequential chain – see Theorem 2(iii).

Remark 2 (The no-macroscopic-layer hypothesis is necessary). *Hypothesis (ii) cannot be dropped when layers have genuine width: if a single antichain layer carries $w_{j_0} = \lceil N/2 \rceil$ and the rest of the weight is spread thinly across the remaining layers, $\|w\|_2^2/N^2 \rightarrow \frac{1}{4}$ and $\rho_{\text{node}}(L) \rightarrow 8/3 > 2$: the sites of the heavy layer are mutually invisible (none lies in another’s cone, being an antichain), which deletes a positive fraction of the comparable pairs from $D(L)$. If instead the heavy layer – and every other layer – is internally totally ordered, $\rho_{\text{node}} = 2N/(N-1)$ regardless of the heavy layer’s size, by part (i)’s exact identity: universality is unconditional only up to intra-layer order, not layer size alone.*

4 Experimental Validation

All measurements are on real reactive graphs built in the reference implementation, NeuroDSL [1], using the exact routine (`_downstream_nodes`) that a real activation-patching sweep would use to restore a corrupted baseline – not a combinatorial simulation. Every identity below is checked at *zero* numerical tolerance (exact integer equality), not asymptotic convergence, except where a limit is explicitly asymptotic.

E1: Fully sequential chain, exhaustive per-node sweep. On a single chain of $N \approx 4000$ nodes partitioned into L layers under three representative (q , profile) combinations ($q = 0$ and $q = 1$ output-heavy, $q = 2$ input-heavy) drawn from the four Karamata profiles of Theorem 1’s own validation, treating every node as a site gives $D(L) = \binom{N}{2}$ *exactly* in every configuration tested ($L \in \{20, 80, 320\}$), hence $\rho_{\text{node}} = 2N/(N-1) \approx 2.0005$ throughout, showing no dependence on L at fixed N – consistent with the $\Theta(1/N)$ rate of Theorem 2(iii), not $\Theta(1/L)$ – where Theorem 1 would predict, under its own depth-stratified sampling protocol, values ranging from 1.5 ($q = 1$, output-heavy) to 4.0 ($q = 2$, input-heavy) on the very same graphs.

E2: Genuine per-layer weighting. Using the real per-layer downstream cones $T_i = \sum_{j > i} w_j$ measured directly on the layered graphs of Theorem 1’s own validation, the identity $2 \sum_i w_i T_i = N^2 - \|w\|_2^2$ holds exactly at every $L \in \{20, 40, 80, 160, 320\}$ tested, for $q \in \{0, 0.5, 1, 2\}$ under both profiles. $\rho_{\text{node}}(L)$ converges to 2 from above, and the measured rate $L(\rho_{\text{node}} - 2)/2$ tracks the predicted $(q+1)^2/(2q+1)$ increasingly closely as L grows (e.g. $q = 2$: predicted rate 1.8, measured 1.778 at $L = 320$), *identically* under both output- and input-heavy orientations at every L – direct confirmation of property (iv), in sharp contrast with Theorem 1’s asymmetric limits on the same profiles.

E3: Necessity of the hypothesis. A layered graph with one artificially macroscopic layer (holding half the total node count) and the remainder spread thinly gives, at the same zero-

tolerance identity check, $\rho_{\text{node}}(L) \rightarrow 8/3 \approx 2.667$, not 2 (measured: 2.7701 at $L = 10$, decreasing monotonically to 2.6892 at $L = 80$, consistent with $8/3 \approx 2.6667$) – confirming Remark 2 on a real graph, not merely as an algebraic possibility.

5 Discussion

This result sharpens, but does not retract, Theorem 1: both theorems are correct statements about the *same* engine, differing only in which sweep protocol they model. A depth-stratified sweep is the right model whenever a practitioner probes a fixed representative site per layer – the common case for architecture-level interpretability protocols that patch, say, one residual-stream position per block. An exhaustive per-node sweep is the right model whenever every individual computable unit is itself a candidate site – neuron-level or edge-level ablation studies, or an automated circuit-discovery search that does not restrict itself to one site per depth. Practitioners should read Theorem 1’s q -dependent ceiling as a property of *coarse-grained* sweeps, and Theorem 2’s universal 2 as the limit that a sufficiently fine-grained exhaustive sweep converges to regardless of where computation is concentrated in the network. The wall-clock refinement of [3] (a ceiling strictly below the combinatorial limit, due to per-call interpreter overhead) applies to either theorem in the same way, since it is orthogonal to the site-sampling design; we do not restate it here.

6 Limitations

This note inherits every limitation of the layered cost model already discussed in [3] (Float32/CPU measurements, the idealized layered abstraction, the site-independent-remainder hypothesis). Theorem 2(ii)’s hypothesis is verified directly under the Karamata profiles tested here, but is not itself tested against a profile that violates Karamata regularity while still satisfying $\max_j w_j/N \rightarrow 0$; we expect the theorem to hold in that broader setting (the proof of (ii) uses only $\max_j w_j/N \rightarrow 0$, not regular variation), but have not constructed such a profile experimentally.

7 Conclusion

The aggregate speedup of an exhaustive sweep on a reactive computational graph is not a single number, and not because the underlying engine is unpredictable: it is a readout of the site-sampling design crossed with the network’s cost-by-depth profile. Depth-stratified sampling gives the q -dependent family of [3]; genuinely exhaustive, per-node sampling collapses that family to a universal 2, via an exact pair-counting identity that holds at every finite depth and requires no assumption on the cost profile beyond the absence of a single dominating layer. Both are theorems about the same locality guarantee, read through two different – and equally realistic – notions of what "exhaustive" means.

References

- [1] Khemais, A. (2026). NeuroDSL: A Dynamic Computational Graph Framework with Optimal Memory Planning. *Preprint*.
- [2] Khemais, A. (2026). Exact Network Surgery: Functional Invariance and Gradient Plasticity in Reactive Computational Graphs. *arXiv preprint arXiv:2607.16568*.
- [3] Khemais, A. (2026). Cost Accounting for Reactive Computational Graphs: Exhaustive Sweeps, Sequential Mutation, and the Backward-Locality Gap. *arXiv preprint arXiv:2607.18323*.